

Business Recommendations

Based on the fraud analytics performed in this project, the following recommendations can improve fraud monitoring and reduce financial risk.

1. Real-Time Fraud Monitoring

Implement real-time monitoring for transactions that exceed predefined risk thresholds or display unusual customer behavior.

2. Customer Risk Profiling

Develop customer risk profiles using historical transaction behavior to identify accounts requiring additional verification.

3. Geographic Monitoring

Flag transactions originating from unfamiliar or high-risk locations and monitor rapid changes in customer transaction locations.

4. Time-Based Fraud Detection

Increase monitoring during periods that consistently show elevated fraud activity.

5. Transaction Pattern Analysis

Identify rapid transaction sequences, unusually large payments, and repeated transaction attempts that may indicate fraudulent behavior.

6. Merchant Risk Assessment

Regularly review merchant categories associated with higher fraud rates and strengthen monitoring for high-risk sectors.

7. Executive Reporting

Maintain interactive dashboards that provide decision-makers with key fraud metrics, trends, and risk indicators to support timely business decisions.